INCH-POUND

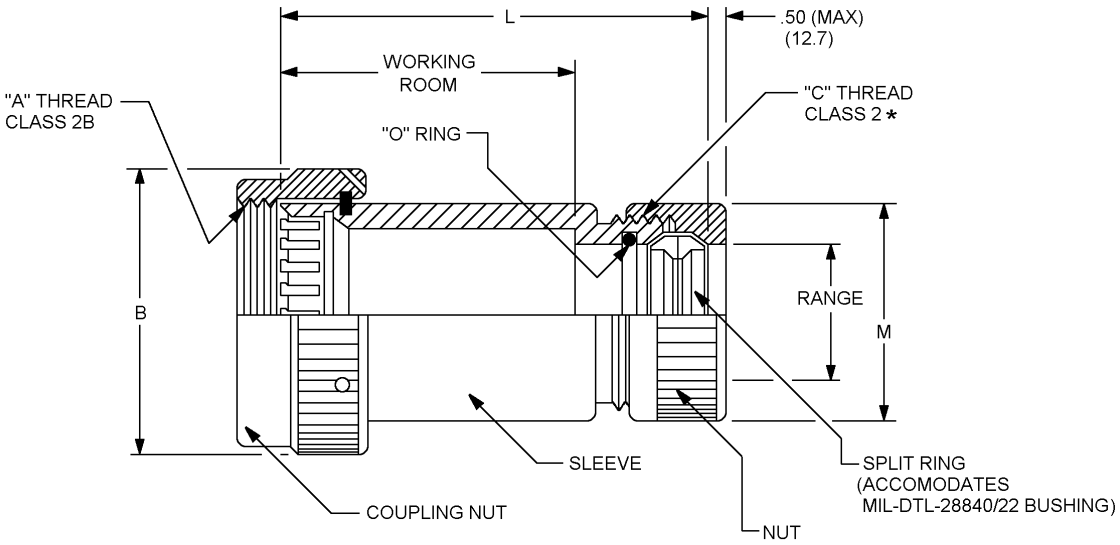
MIL-DTL-28840/5B
20 April 2004
SUPERSEDING
MIL-C-28840/5A(ES)
26 May 1981

DETAIL SPECIFICATION SHEET

# CONNECTORS, ELECTRICAL, CIRCULAR, THREADED, HIGH SHOCK, HIGH DENSITY, SHIPBOARD, BACKSHELL, STRAIGHT, METAL CONDUIT FOR EMI SHIELDING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-28840.

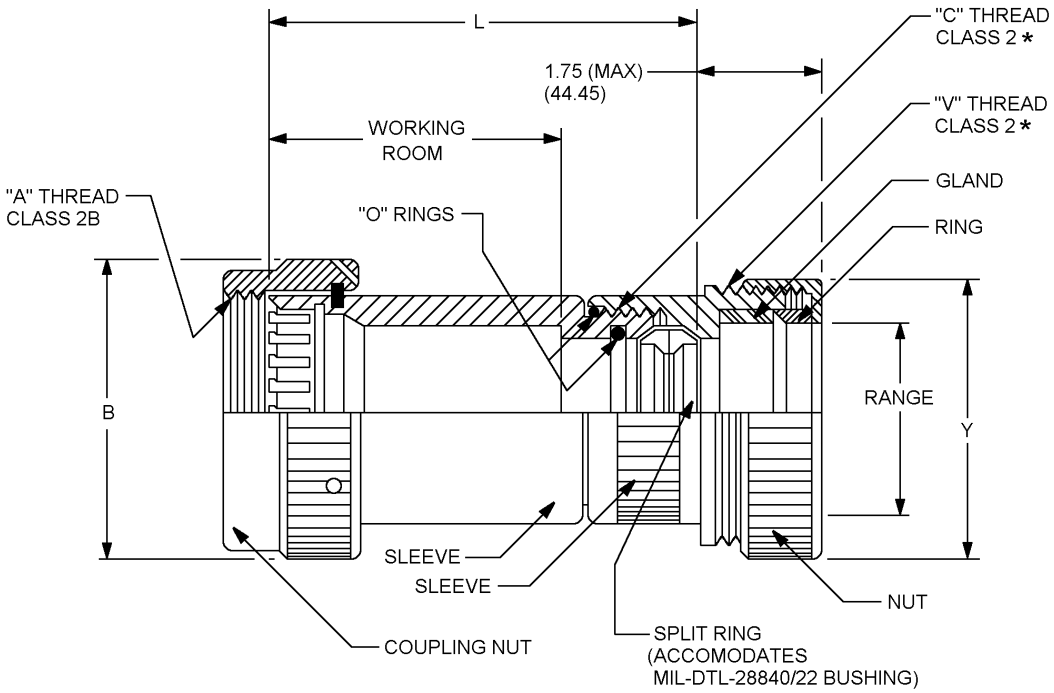



STYLE A
(FOR MIL-DTL-28840/4-**A PLAIN CONDUIT)

FIGURE 1. Dimensions and configurations.



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STYLE B

(FOR MIL-DTL-28840/4-**B JACKET CONDUIT)

FIGURE 1. Dimensions and configurations - Continued.



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| Dash no. | Designator (see note 1) | A thread class 2B | B max dia | C thread class 2* | Sealing range | |
|---|---|---|---|---|---|---|
| | | | | | Max | Min |
| 01 | A (11) | .750-20 UNEF | 1.028 (26.11) | .875-20 UNEF | .63 (16.0) | .50 (12.7) |
| 02 | B (13) | .875-20 UNEF | 1.141 (28.98) | | | |
| 03 | | | | 1.000-20 UNEF | .75 (19.1) | .63 (16.0) |
| 04 | C (15) | 1.000-20 UNEF | 1.263 (32.08) | .875-20 UNEF | .63 (16.0) | .50 (12.7) |
| 05 | | | | 1.000-20 UNEF | .75 (19.1) | .63 (16.0) |
| 06 | | | | 1.125-18 UNEF | .88 (22.4) | .75 (19.1) |
| 07 | | | | 1.250-18 UNEF | 1.00 (25.4) | .88 (22.4) |
| 08 | D (17) | 1.125-18 UNEF | 1.387 (35.23) | 1.125-18 UNEF | .88 (22.4) | .75 (19.1) |
| 09 | | | | 1.250-18 UNEF | 1.00 (25.4) | .88 (22.4) |
| 10 | | | | 1.625-18 UNEF | 1.38 (35.1) | 1.19 (30.2) |
| 11 | E (19) | 1.250-18 UNEF | 1.513 (38.43) | 1.125-18 UNEF | .88 (22.4) | .75 (19.1) |
| 12 | | | | 1.250-18 UNEF | 1.00 (25.4) | .88 (22.4) |
| 13 | | | | 1.625-18 UNEF | 1.38 (35.1) | 1.19 (30.2) |
| 14 | F (23) | 1.438-18 UNEF | 1.703 (45.51) | 1.250-18 UNEF | 1.00 (25.4) | .88 (22.4) |
| 15 | | | | 1.625-18 UNEF | 1.38 (35.1) | 1.19 (30.2) |
| 16 | | | | 1.875-18 UNS | 1.63 (41.4) | 1.44 (36.6) |
| 17 | G (25) | 1.562-18 UNEF | 1.825 (46.35) | 1.250-18 UNEF | 1.00 (25.4) | .88 (22.4) |
| 18 | | | | 1.625-18 UNEF | 1.38 (35.1) | 1.19 (30.2) |
| 19 | | | | 1.875-18 UNS | 1.63 (41.4) | 1.44 (36.6) |
| 20 | | | | 2.125-16 UN | 1.88 (47.8) | 1.63 (41.4) |
| 21 | H (29) | 1.875-16 UN | 2.143 (54.43) | 1.875-18 UNS | 1.63 (41.4) | 1.44 (36.6) |
| 22 | | | | 2.125-16 UN | 1.88 (67.8) | 1.63 (41.4) |
| 23 | J (33) | 2.062-16 UNS | 2.329 (59.16) | 2.125-16 UN | 1.88 (47.8) | 1.63 (41.4) |
| 24 | | | | 2.750-16 UN | 2.38 (60.5) | 2.13 (54.1) |
| 25 | A (11) | .750-20 UNEF | 1.028 (26.11) | .875-20 UNEF | .75 (19.05) | .63 (16.00) |

See note at end of figure.

FIGURE 1. Dimensions and configurations - Continued.



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| Dash no. | Designator (see note 1) | Conduit size | L ± .12 (± 3.05) | Working room ref. | V thread class 2* | M max dia | Y max dia |
|---|---|---|---|---|---|---|---|
| 01 | A (11) | 03 | 1.75 (44.45) | 1.00 (25.4) | 1.000-20 UNEF | 1.12 (28.45) | 1.19 (30.23) |
| 02 | B (13) | | | | | | |
| 03 | | 04 | | | 1.188-18 UNEF | 1.25 (31.8) | 1.38 (35.1) |
| 04 | C (15) | 03 | | | 1.000-20 UNEF | 1.12 (28.5) | 1.19 (30.23) |
| 05 | | 04 | | | 1.188-30 UNEF | 1.25 (31.8) | 1.38 (35.1) |
| 06 | | 05 | | | 1.438-18 UNEF | 1.43 (36.3) | 1.63 (41.4) |
| 07 | | 06 | 3.00 (76.2) | | 1.750-18 UNS | 1.62 (41.2) | 1.94 (49.3) |
| 08 | D (17) | 05 | 2.25 (57.15) | 1.50 (38.10) | 1.438-18 UNEF | 1.43 (36.3) | 1.63 (41.4) |
| 09 | | 06 | | | 1.750-18 UNS | 1. 62 (41.2) | 1.94 (49.3) |
| 10 | | 08 | | 2.00 (50.80) | 2.00-18 UNS | 1.81 (46.0) | 2.19 (55.6) |
| 11 | (E19) | 05 | | 1.50 (38.10) | 1.438-18 UNEF | 1.43 (36.3) | 1.63 (41.4) |
| 12 | | 06 | | | 1.750-18 UNEF | 1.62 (41.2) | 1.94 (49.3) |
| 13 | | 08 | | | 2.00-18 UNS | 1.81 (46.2) | 2.19 (55.6) |
| 14 | F (23) | 06 | 2.75 (69.85) | 2.00 (50.80) | 1.750-18 UNS | 1.62 (41.2) | 1.94 (49.3) |
| 15 | | 08 | | | 2.00-18 UN | 1.81 (46.0) | 2.19 (55.6) |
| 16 | | 10 | 3.50 (88.90) | | 2.00-16 UN | 2.18 (55.4) | 2.44 (62.0) |
| 17 | G (25) | 06 | 2.75 (60.85) | | 1.750-18 UNS | 1.62 (41.2) | 1.94 (49.3) |
| 18 | | 08 | | | 2.00-18 UN | 1.81 (46.0) | 2.19 (55.6) |
| 19 | | 10 | | | 2.250-16 UN | 2.18 (55.4) | 2.44 (62.0) |
| 20 | | 12 | 4.00 (101.6) | 2.50 (63.50) | 2.500-16 UN | 2.56 (65.0) | 2.69 (68.3) |
| 21 | H (29) | 10 | 3.25 (82.55) | | 2.250-16 UN | 2.18 (55.4) | 2.44 (62.0) |
| 22 | | 12 | | | 2.500-16 UN | 2.56 (63.0) | 2.66 (68.3) |
| 23 | J (33) | 12 | | | | 2.56 (65.0) | 2.60 (88.3) |
| 24 | | 16 | | | 3.00-16 UN | 3.06 (77.8) | 3.18 (81.0) |
| 25 | A (11) | 04 | 1.75 (44.45) | 1.00 (25.4) | 1.188-18 UNEF | 1.75 (31.75) | 1.38 (95.05) |

NOTES:
1. Shell sizes are provided within parentheses for information and are not a part of the designator.
2. Dimensions are in inches. Dimensions apply after plating.
3. Metric equivalents are given for general information only.
4. Dimensions apply after plating.



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FIGURE 1. Dimensions and configurations - Continued.

REQUIREMENTS

Dimensions and configurations: See figure 1.

Interface dimensions in accordance with MIL-DTL-28840.

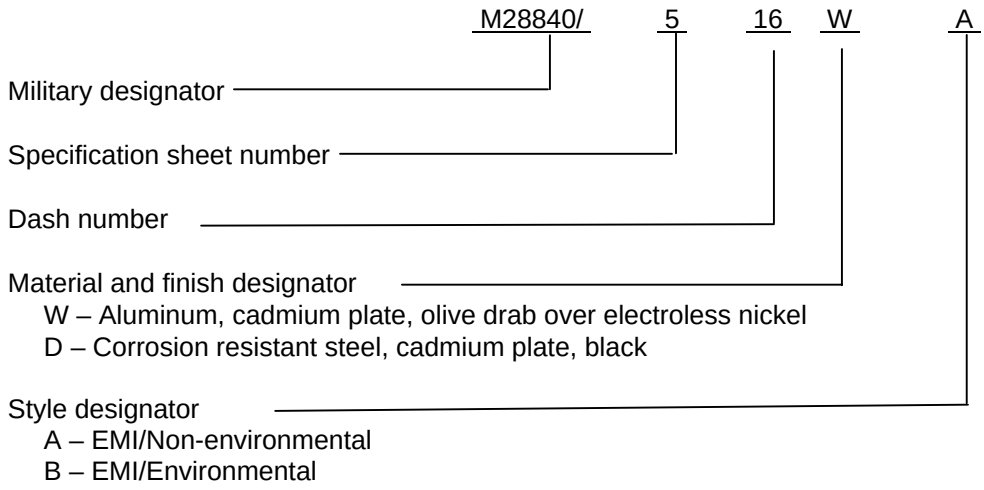
Material: Aluminum alloy in accordance with SAE-AMS-QQ-A-225, ASTM B85, ASTM B26/B26M, or ASTM B108. Corrosion resistant steel in accordance with SAE-AMS-QQ-S-763, class 316.

Finish: Aluminum components shall be cadmium plate, olive drab over electroless nickel.

Corrosion resistant steel components shall be cadmium plate, black. The screws and washers shall be passivate in accordance with SAE-AMS-QQ-P-35.

Marking: The backshell shall be identified with the manufacturer's name or trademark and the Part or Identifying Number (PIN).

PIN example:




Qualification: Two samples in each shell size for which qualification is desired, shall be tested in accordance with MIL-DTL-28840.

Qualification of MIL-DTL-28840/6 will qualify MIL-DTL-28840/5 by similarity for group 5 tests.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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Reference documents. In addition to MIL-DTL-28840, this document references the following:

| | | |
|---|---|---|
| ASTM B26/B26M | SAE-AMS-QQ-P-35 | MIL-DTL-28840/4 |
| ASTM B85 | SAE-AMS-QQ-A-225 | MIL-DTL-28840/6 |
| ASTM B108 | SAE-AMS-QQ-S-763 | MIL-DTL-28840/22 |

## CONCLUDING MATERIAL

Custodians:
Navy - EC
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4416-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online data at http://www.dodssp.daps.mil.



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